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Branching Stochastic Neural Networks: A Count-Valued Recurrent Network View of Fission Multiplication

Draft manuscript

Author name(s) to be added

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Abstract

Nuclear fission multiplication is conventionally modeled by neutron transport, branching processes, and eigenvalue criticality theory. This manuscript develops a complementary neural-network formulation that is mathematically precise but intentionally limited in scope. We define a *branching stochastic neural network* (BSNN), a count-valued stochastic recurrent network whose transition law satisfies a probability-generating-function factorization. We prove that this factorization is equivalent to the independent-offspring property of a multitype branching process, and we extend the characterization to continuous state spaces by proving an equivalent branching-property theorem for Markov kernels on finite counting measures. Under the usual generation-based abstraction, discretized fission multiplication is therefore an exact instance of a BSNN. Its expected dynamics reduce to the positive linear recurrent neural network

$$x_{n+1} = Mx_n,$$

where the nonnegative matrix M is the next-generation mean-offspring operator. We derive mean and covariance propagation, identify the asymptotic multiplication factor with the Perron–Frobenius spectral radius $\rho(M)$, state the associated extinction criterion, and show that backpropagation through the mean BSNN is algebraically identical to an adjoint sensitivity calculation. We add finite-sample guarantees for estimating M and offspring probability-generating functions, propose nonnegative recurrent architectures with extinction, covariance, and spectral-radius regularization, and report synthetic cascade and adjoint-verification benchmarks. We also prove expressivity limitations: the mean dynamics are positive, linear, homogeneous, and monotone, and the stochastic law is constrained by branching factorization. Thus fission multiplication should not be described as a general artificial neural network. Rather, it is rigorously representable as a positive, count-valued, stochastic recurrent computation graph whose physical interpretation is a branching transport process.

Keywords: branching process; stochastic neural network; recurrent neural network; Perron–Frobenius theory; neutron multiplication; fission chains; adjoint sensitivity; positive systems.

1 Introduction

The phrase “nuclear fission is a neural network” is not correct if interpreted literally. A fission chain does not learn by backpropagation, does not contain trainable semantic parameters, and does not implement the broad function classes associated with modern artificial neural networks. The established mathematical language of neutron multiplication is instead neutron transport, branching processes, and positive eigenvalue theory. Modern work on stochastic neutron transport explicitly connects neutron transport with non-local branching Markov processes [11], and stochastic formulations of the neutron transport equation relate its mean semigroup to branching Markov processes and Perron–Frobenius asymptotics [10]. Recent fission-chain work also uses branching-process theory for probabilistic neutron processes in multiplying systems [14].

Nevertheless, there is a restricted sense in which the neural-network viewpoint is exact and potentially useful. A generation-indexed fission chain is a layered stochastic computation graph. In a finite type-space discretization, the expected generation-to-generation map is

$$x_{n+1} = Mx_n, \quad (1)$$

with $M \geq 0$. Equation (1) is exactly a linear recurrent neural network (RNN) with identity activation, zero bias, and nonnegative weights. At the event level, however, fission multiplication is not merely a deterministic linear RNN: the population is count-valued, the width of the graph is random, and extinction and fluctuations matter. The more faithful neural abstraction is therefore a *branching stochastic neural network*.

The goal of this manuscript is to turn the informal analogy into a precise mathematical statement. We ask:

Which stochastic recurrent neural networks are exactly multitype branching processes, and in what sense does discretized fission multiplication belong to that class?

The answer is a law-level factorization theorem. A count-valued stochastic recurrent network is branching if and only if its transition probability-generating functions satisfy

$$G_{x+y}(z) = G_x(z)G_y(z), \quad (2)$$

for all population states $x, y \in \mathbb{N}_0^m$. This identity is stronger than a resemblance between equations; it characterizes the transition law itself.

Contributions

This draft makes the following contributions.

1. It defines branching stochastic neural networks (BSNNs) as count-valued stochastic recurrent networks satisfying the generating-function factorization (2).
2. It proves an if-and-only-if equivalence between finite-type BSNN transition laws and multitype branching-process offspring laws.
3. It proves a continuous-state BSNN characterization for branching Markov kernels on finite counting measures over a standard Borel state space.

4. It shows that a generation-discretized fission multiplication model is an exact BSNN under the standard independent-branching approximation.
5. It derives mean and covariance recursions, showing that the mean model is a positive linear RNN while the full process retains branching fluctuations.
6. It gives finite-sample estimation guarantees for the mean matrix and for offspring PGFs under parent-resolved observations, and a regression guarantee for aggregate count trajectories.
7. It connects criticality to the Perron–Frobenius spectral radius, extinction to fixed points of offspring generating functions, and gradient backpropagation to adjoint sensitivity analysis.
8. It develops nonnegative recurrent architectures with explicit spectral-radius, extinction, covariance, and branching-structure regularization, and compares them with an unconstrained RNN on synthetic cascade data.
9. It proves expressivity limitations, preventing the analogy from being overstated.

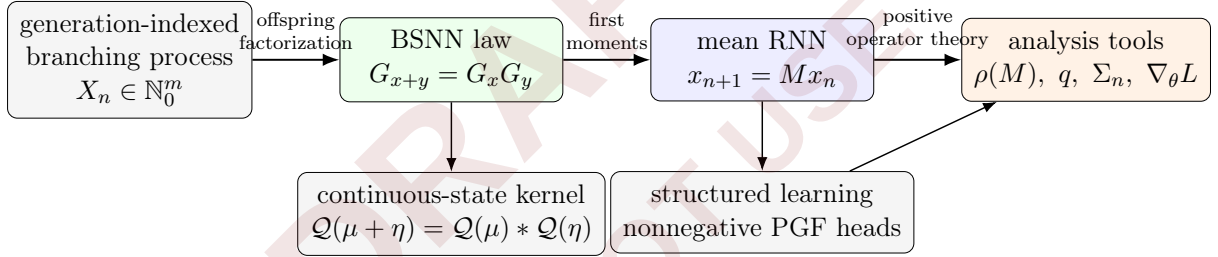


Figure 1: Schematic of the manuscript’s main logic. The physical branching abstraction implies a BSNN law-level factorization. Taking first moments gives a positive linear recurrent network. The resulting positive operator supports criticality, extinction, covariance, and adjoint-sensitivity analyses.

Scope and safety note

The paper is a mathematical abstraction. The “types” below are generic labels in a finite state space. No material compositions, cross sections, device geometries, enrichment levels, control protocols, or operational design parameters are specified or required. The numerical section proposes synthetic positive-matrix examples only. The intended contribution is to stochastic-process theory, positive recurrent computation graphs, and differentiable sensitivity analysis, not to the design or operation of nuclear systems.

2 Related work

Branching and neutron transport. Classical branching processes are treated in depth by Harris [7] and Athreya and Ney [1]. In neutron transport, recent probability-theoretic work has developed stochastic representations of the neutron transport equation using branching Markov processes and positive semigroup theory [3, 4, 8, 10, 11]. These works are mathematically closer to fission multiplication than generic neural-network

theory. The present manuscript does not attempt to replace those models. Instead, it recasts their finite-generation, finite-type abstraction as a particular stochastic recurrent network class.

Machine learning in reactor physics. Neural networks have been used as solvers, surrogates, and accelerators for reactor-physics calculations. Examples include physics-informed neural networks for neutron diffusion and reactor calculations [5], data-enabled PINNs for neutron diffusion eigenvalue problems [17], LSTM-based transport calculations [13], and recent neural-operator surrogate models for neutron transport [16]. A 2025 review summarizes broader AI applications in reactor physics and identifies theoretical and generalization challenges [18]. The present work asks a different question: not whether neural networks can solve neutron-physics equations, but whether the physical branching law itself defines a mathematically recognizable stochastic neural network.

Statistical inference and adjoints. Statistical inference for branching processes has a long history; recent work continues to study sampling distributions and estimators for multitype branching processes under family-size sampling [20]. Our finite-sample bounds are elementary concentration guarantees tailored to the BSNN formulation rather than a replacement for that literature. The adjoint part of the manuscript is also closely related to classical sensitivity analysis [19]. Recent reactor-physics work has demonstrated computational equivalence between continuous and discrete adjoint methods for time-dependent multigroup neutron diffusion models [24]. Here we give a small discrete benchmark showing that reverse-mode automatic differentiation through a discretized positive operator returns exactly the same vector-Jacobian product as the classical discrete adjoint solve.

Positive systems and Perron–Frobenius theory. The spectral-radius characterization of growth is an instance of Perron–Frobenius theory for nonnegative matrices and positive operators [2, 9]. In continuous spaces, related positive-operator results are usually formulated through Krein–Rutman theory [12]. Neural-network language adds an interpretation of repeated positive multiplication as recurrent signal propagation, but the underlying mathematics is positive linear systems theory.

3 Notation and preliminaries

Let $m \in \mathbb{N}$ be the number of types. A type may represent an abstract neutron state, a spatial cell, an energy group, an angular bin, a delayed-family label, or any other finite state label. We write

$$X_n = (X_{n,1}, \dots, X_{n,m})^\top \in \mathbb{N}_0^m$$

for the random count vector at generation n . For $z \in [0, 1]^m$ and $y \in \mathbb{N}_0^m$, define

$$z^y = \prod_{j=1}^m z_j^{y_j}.$$

For a transition law P_x from current state x to next state X_{n+1} , its probability-generating function (PGF) is

$$G_x(z) = \mathbb{E} [z^{X_{n+1}} \mid X_n = x]. \quad (3)$$

PGFs uniquely determine probability laws on $\mathbb{N}_0^m$.

We use the convention that a matrix $M \in \mathbb{R}^{m \times m}$ acts on column population vectors by

$$x_{n+1} = Mx_n.$$

The entry M_{ji} is the expected number of type- j next-generation units produced by one type- i present-generation unit.

4 Branching stochastic neural networks

Definition 4.1 (Count-valued stochastic recurrent network). *A count-valued stochastic recurrent network is a time-homogeneous Markov chain $(X_n)_{n \geq 0}$ on $\mathbb{N}_0^m$ with transition kernels $P_x(\cdot)$ and transition PGFs G_x as in (3). The state $X_{n,i}$ is interpreted as the activation count of type- i units at layer or generation n .*

Definition 4.2 (Branching stochastic neural network). *A count-valued stochastic recurrent network is a branching stochastic neural network, abbreviated BSNN, if*

$$G_{x+y}(z) = G_x(z)G_y(z) \quad (4)$$

for all $x, y \in \mathbb{N}_0^m$ and all $z \in [0, 1]^m$, with $G_0(z) = 1$.

The factorization property says that the next-layer output distribution from a sum of independent input populations is the convolution of the output distributions from the parts. It is the law-level version of the branching property. Figure 2 depicts this convolution/product identity.

Theorem 4.3 (Law-level characterization of BSNNs). *Let (X_n) be a count-valued stochastic recurrent network on $\mathbb{N}_0^m$ with transition PGFs $(G_x)_{x \in \mathbb{N}_0^m}$. The following are equivalent.*

(i) *The PGFs satisfy the branching factorization (4).*

(ii) *There exist random vectors $Y_i \in \mathbb{N}_0^m$, $i = 1, \dots, m$, such that, conditionally on $X_n = x$,*

$$X_{n+1} \stackrel{d}{=} \sum_{i=1}^m \sum_{r=1}^{x_i} Y_{i,r}^{(n)}, \quad (5)$$

where $Y_{i,r}^{(n)}$ are mutually independent copies of Y_i .

Proof. Assume first that (5) holds. Let

$$g_i(z) = \mathbb{E}[z^{Y_i}].$$

Conditionally on $X_n = x$, independence of the offspring vectors gives

$$\begin{aligned} G_x(z) &= \mathbb{E} \left[z^{\sum_{i=1}^m \sum_{r=1}^{x_i} Y_{i,r}^{(n)}} \right] \\ &= \prod_{i=1}^m g_i(z)^{x_i}. \end{aligned}$$

Therefore

$$G_{x+y}(z) = \prod_{i=1}^m g_i(z)^{x_i+y_i} = \left(\prod_{i=1}^m g_i(z)^{x_i} \right) \left(\prod_{i=1}^m g_i(z)^{y_i} \right) = G_x(z)G_y(z).$$

Thus (4) holds.

Conversely, suppose (4) holds. Let e_i denote the i th coordinate vector and define

$$g_i(z) = G_{e_i}(z).$$

For any $x \in \mathbb{N}_0^m$,

$$x = \sum_{i=1}^m x_i e_i.$$

Repeated application of (4) yields

$$G_x(z) = \prod_{i=1}^m G_{e_i}(z)^{x_i} = \prod_{i=1}^m g_i(z)^{x_i}. \quad (6)$$

Because each g_i is the PGF of the transition law from state e_i , there exists an $\mathbb{N}_0^m$ -valued random vector Y_i with PGF g_i . If $Y_{i,r}^{(n)}$ are independent copies of Y_i , then the PGF of the sum on the right-hand side of (5) is exactly (6). Since PGFs uniquely determine laws on $\mathbb{N}_0^m$, the transition law equals the law of the independent-offspring sum. This proves the equivalence. $\square$

Remark 4.4 (Why this is stronger than a metaphor). A *deterministic linear recurrence* $x_{n+1} = Mx_n$ only describes first moments. Theorem 4.3 characterizes the entire transition law. Thus the BSNN formulation is not merely a relabeling of a matrix equation; it identifies exactly which count-valued stochastic recurrent networks have the branching property.

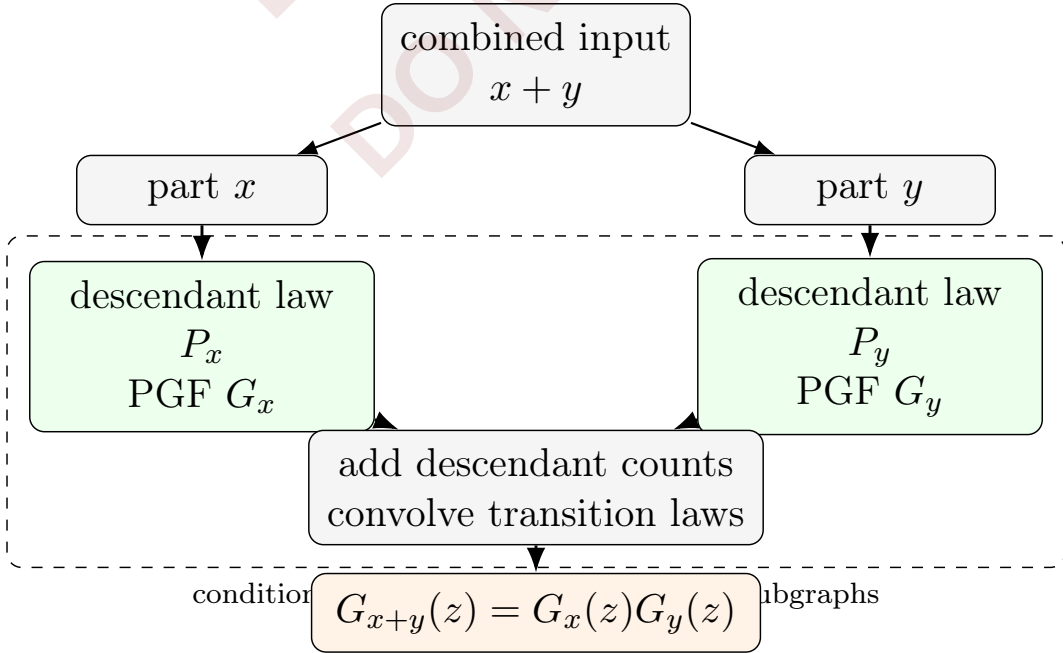


Figure 2: Law-level branching factorization. The transition law from a superposed population $x + y$ is the convolution of the independent descendant laws from x and y . PGFs convert this convolution statement into the product identity $G_{x+y} = G_x G_y$.

5 Fission multiplication as a BSNN

In a generation-based fission-chain abstraction, each present-generation neutron or event produces a random vector of next-generation descendants. Let $Y_i \in \mathbb{N}_0^m$ denote the random descendant-type vector produced by one type- i unit. Then

$$X_{n+1} = \sum_{i=1}^m \sum_{r=1}^{X_{n,i}} Y_{i,r}^{(n)}. \quad (7)$$

This is precisely the multitype branching-process form. In physical neutron transport, the state description may be continuous, and the discrete-type model should be understood as a coarse graining or discretization of a transport process.

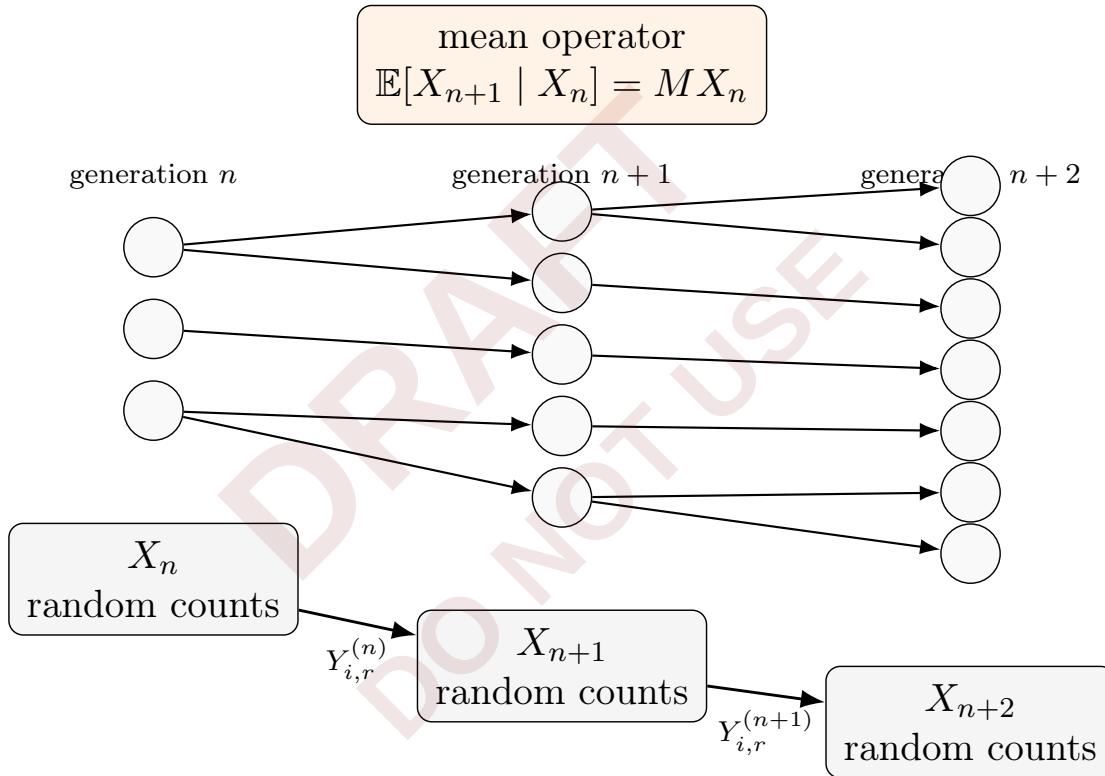


Figure 3: A realized BSNN or fission-chain genealogy is a layered random computation graph. Each active unit produces a random number and type of descendants. Averaging over the random offspring graph yields the deterministic positive recurrence $x_{n+1} = Mx_n$.

Proposition 5.1 (Discretized fission multiplication is a BSNN). *Under a finite-type, generation-indexed independent-offspring approximation, fission multiplication has transition law (7). Hence it is a branching stochastic neural network in the sense of Definition 4.2.*

Proof. Equation (7) is exactly the independent-offspring representation (5). The result follows immediately from Theorem 4.3. $\square$

The neural-network dictionary is shown in Table 1. This dictionary is algebraic and stochastic; it does not imply biological cognition or machine learning inside the physical system.

Table 1: A finite-type fission-chain abstraction as a stochastic recurrent network.

Neural-network term	Fission-chain abstraction
Layer or recurrent step	Neutron generation
Unit type	Discretized neutron/event state
Activation	Count of units of a given type
Weight M_{ji}	Mean number of type- j descendants from type i
Activation function	Identity in the mean model; offspring law in the stochastic model
Hidden state X_n or x_n	Random or mean population vector
Gain/stability parameter	Spectral radius $\rho(M)$, analogous to multiplication factor
Computation graph	Random layered genealogy

6 Mean and covariance propagation

Assume each offspring vector Y_i has finite second moments. Define

$$M_{ji} = \mathbb{E}[(Y_i)_j], \quad V_i = \text{Cov}(Y_i) \in \mathbb{R}^{m \times m}. \quad (8)$$

Let

$$x_n = \mathbb{E}[X_n], \quad \Sigma_n = \text{Cov}(X_n).$$

Theorem 6.1 (Moment propagation). *For a BSNN with finite second moments,*

$$\mathbb{E}[X_{n+1} \mid X_n] = MX_n, \quad (9)$$

$$x_{n+1} = Mx_n, \quad (10)$$

$$\Sigma_{n+1} = M\Sigma_n M^\top + \sum_{i=1}^m x_{n,i} V_i. \quad (11)$$

Proof. The j th coordinate of X_{n+1} is

$$X_{n+1,j} = \sum_{i=1}^m \sum_{r=1}^{X_{n,i}} (Y_{i,r}^{(n)})_j.$$

Conditioning on X_n and using the definition of M_{ji} gives

$$\mathbb{E}[X_{n+1,j} \mid X_n] = \sum_{i=1}^m X_{n,i} M_{ji},$$

which is the j th coordinate of MX_n . This proves (9). Taking expectation gives

$$x_{n+1} = \mathbb{E}[\mathbb{E}[X_{n+1} \mid X_n]] = \mathbb{E}[MX_n] = Mx_n,$$

which proves (10).

For covariance, conditional independence of offspring from distinct present-generation units gives

$$\text{Cov}(X_{n+1} \mid X_n) = \sum_{i=1}^m X_{n,i} V_i.$$

The law of total covariance gives

$$\begin{aligned}\Sigma_{n+1} &= \mathbb{E}[\text{Cov}(X_{n+1} \mid X_n)] + \text{Cov}(\mathbb{E}[X_{n+1} \mid X_n]) \\ &= \sum_{i=1}^m x_{n,i} V_i + \text{Cov}(M X_n) \\ &= \sum_{i=1}^m x_{n,i} V_i + M \Sigma_n M^\top.\end{aligned}$$

This proves (11). $\square$

Corollary 6.2 (Mean BSNN as a linear recurrent neural network). *The mean dynamics of a BSNN are exactly the recurrent neural network*

$$h_{n+1} = \sigma(W h_n + b) \quad (12)$$

with

$$W = M, \quad b = 0, \quad \sigma = \text{id}. \quad (13)$$

Moreover, W is entrywise nonnegative.

Proof. Substitute $h_n = x_n$, $W = M$, $b = 0$, and $\sigma = \text{id}$ in the standard RNN update. The result is (10). Nonnegativity follows from $M_{ji} = \mathbb{E}[(Y_i)_j]$. $\square$

Remark 6.3 (Why covariance matters). *The deterministic recurrence (10) hides extinction probabilities, fluctuations, and correlations. Equation (11) shows how stochasticity enters as an additive branching-noise term. This distinguishes a BSNN from an ordinary deterministic linear RNN.*

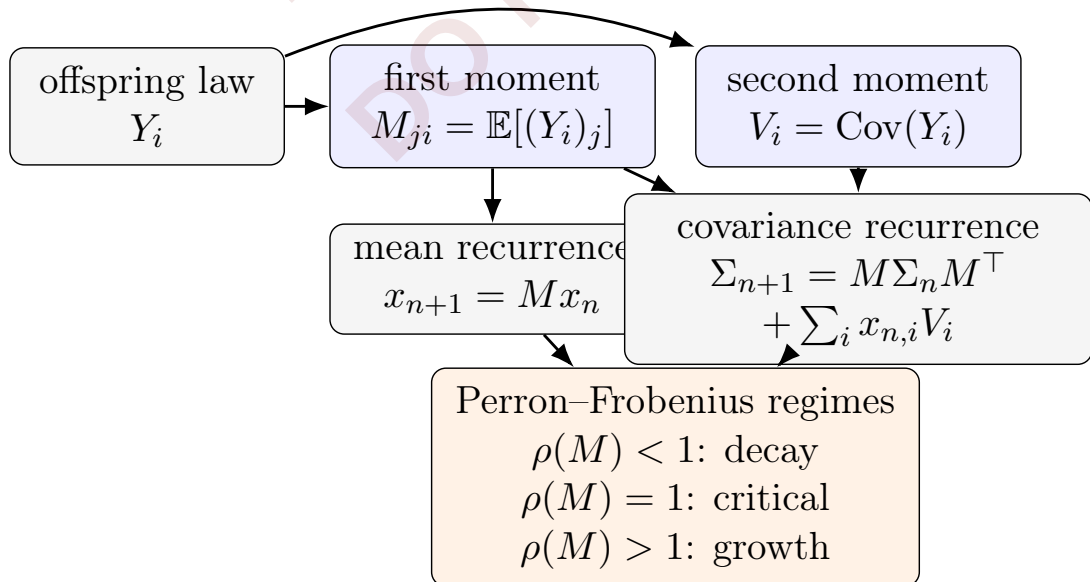


Figure 4: Moment and criticality structure. First moments give the positive linear RNN, second moments add the branching-noise term, and the Perron–Frobenius spectral radius controls the asymptotic mean growth regime.

7 Criticality, spectral radius, and extinction

The matrix M is the next-generation mean-offspring matrix. The asymptotic behavior of the mean population is governed by its Perron–Frobenius spectral radius.

Theorem 7.1 (Spectral-radius criticality). *Let $M \in \mathbb{R}^{m \times m}$ be nonnegative and primitive. Let*

$$k = \rho(M),$$

where $\rho(M)$ is the spectral radius. For any nonzero $x_0 \geq 0$,

$$x_n = M^n x_0 \tag{14}$$

satisfies

$$\lim_{n \rightarrow \infty} \|x_n\|^{1/n} = k. \tag{15}$$

Consequently, $k < 1$ implies $x_n \rightarrow 0$; if $k = 1$, then x_n converges to the positive Perron-mode multiple $(vu^\top x_0)/(u^\top v)$; and if $k > 1$, then the mean population grows exponentially at rate k .

Proof. By the Perron–Frobenius theorem for primitive nonnegative matrices, there exist strictly positive right and left eigenvectors $v > 0$ and $u > 0$ such that

$$Mv = kv, \quad u^\top M = ku^\top,$$

and

$$\frac{M^n}{k^n} \rightarrow \frac{vu^\top}{u^\top v}. \tag{16}$$

Since $x_0 \geq 0$ and $x_0 \neq 0$, we have $u^\top x_0 > 0$. Multiplying (16) by x_0 gives

$$M^n x_0 = k^n \left(\frac{vu^\top x_0}{u^\top v} + o(1) \right),$$

where the leading vector is nonzero and positive. Therefore $\|M^n x_0\| \sim Ck^n$ for some $C > 0$, and taking n th roots yields (15). The subcritical, critical, and supercritical conclusions follow immediately. $\square$

Remark 7.2 (Relation to k_{eff}). *In reactor-physics language, the effective multiplication factor is an eigenvalue or asymptotic generation-gain quantity. In the finite-type abstraction of this paper, the corresponding gain is $\rho(M)$. More refined transport models may define the relevant operator and eigenvalue in continuous phase space rather than through a finite matrix.*

Mean growth is not the same as survival. Let

$$g_i(z) = \mathbb{E}[z^{Y_i}], \quad g(z) = (g_1(z), \dots, g_m(z)). \tag{17}$$

Let q_i be the extinction probability starting from one type- i unit, and let $q = (q_1, \dots, q_m)$.

Theorem 7.3 (Extinction fixed point). *For a multitype BSNN, the extinction-probability vector q is the minimal fixed point of*

$$q = g(q) \tag{18}$$

in $[0, 1]^m$. Under standard irreducibility and nondegeneracy assumptions for multitype branching processes,

$$\rho(M) \leq 1 \implies q = \mathbf{1}, \quad (19)$$

whereas

$$\rho(M) > 1 \implies q < \mathbf{1} \quad (20)$$

componentwise.

Proof sketch. Starting from one type- i unit, extinction occurs if all offspring subtrees become extinct. Conditioning on the first-generation offspring vector Y_i gives

$$q_i = \mathbb{E} \left[\prod_{j=1}^m q_j^{(Y_i)_j} \right] = g_i(q).$$

Thus $q = g(q)$. Iterating from $q^{(0)} = 0$ gives the increasing sequence $q^{(n+1)} = g(q^{(n)})$, where $q^{(n)}$ is the probability of extinction by generation n . Hence the limit is the minimal fixed point. The dichotomy in terms of $\rho(M)$ is the standard irreducible multitype Galton–Watson extinction theorem; see, for example, Harris [7] and Athreya and Ney [1]. A one-type proof is included in Appendix A. $\square$

8 Continuous-state BSNNs and branching Markov kernels

The finite-type model is a discretization. A more intrinsic formulation uses finite counting measures on a general state space. Let $(S, \mathcal{S})$ be a standard Borel space and let $\mathcal{N}_f(S)$ denote the set of finite counting measures on S , equipped with its usual evaluation σ -field. A state is

$$\mu = \sum_{r=1}^n \delta_{s_r},$$

where $s_r \in S$ are the current particles or units. For a measurable function $h : S \rightarrow [0, 1]$, define

$$h^\mu = \exp \left(\int_S \log h(s) \mu(ds) \right),$$

with the usual convention that $h^\mu = \prod_{r=1}^n h(s_r)$ for $\mu = \sum_r \delta_{s_r}$ and $h^0 = 1$.

Let $\mathcal{Q}$ be a Markov kernel from $\mathcal{N}_f(S)$ to $\mathcal{N}_f(S)$. Its probability-generating functional is

$$G_\mu[h] = \int_{\mathcal{N}_f(S)} h^\nu \mathcal{Q}(\mu, d\nu). \quad (21)$$

Definition 8.1 (Continuous-state BSNN kernel). *A Markov kernel $\mathcal{Q}$ on $\mathcal{N}_f(S)$ is a continuous-state BSNN kernel if*

$$G_{\mu+\eta}[h] = G_\mu[h] G_\eta[h] \quad (22)$$

for all $\mu, \eta \in \mathcal{N}_f(S)$ and all measurable $h : S \rightarrow [0, 1]$, with $G_0[h] = 1$.

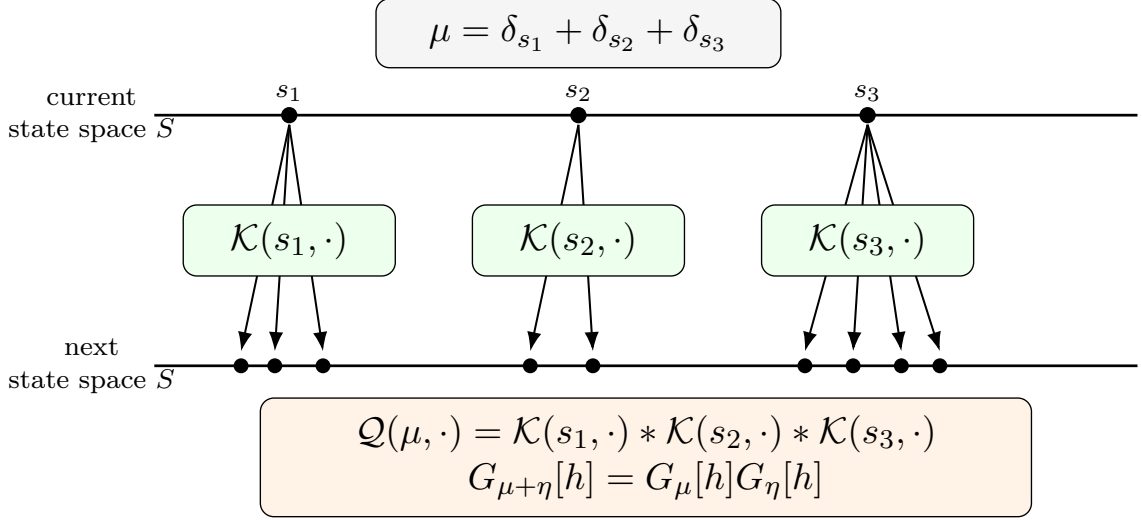


Figure 5: Continuous-state BSNN kernel. The finite vector of type counts is replaced by a finite counting measure on a state space S . Branching means that each atom s evolves through a single-parent kernel $\mathcal{K}(s, \cdot)$ and the offspring measures are added.

Theorem 8.2 (Continuous-state BSNN characterization). *Let $(S, \mathcal{S})$ be standard Borel and let $\mathcal{Q}$ be a Markov kernel on $\mathcal{N}_f(S)$. The following are equivalent.*

- (i) $\mathcal{Q}$ satisfies the branching factorization (22).
- (ii) There exists a measurable single-parent offspring Markov kernel $\mathcal{K} : S \times \mathcal{B}(\mathcal{N}_f(S)) \rightarrow [0, 1]$ such that, for every $\mu = \sum_{r=1}^n \delta_{s_r}$,

$$\mathcal{Q}(\mu, \cdot) = \mathcal{K}(s_1, \cdot) * \cdots * \mathcal{K}(s_n, \cdot), \quad (23)$$

where $*$ denotes convolution under addition of counting measures. Equivalently, conditionally on $X_n = \mu$, the next state has the representation

$$X_{n+1} \stackrel{d}{=} \sum_{r=1}^n Z_r, \quad Z_r \sim \mathcal{K}(s_r, \cdot) \quad (24)$$

with $Z_1, \dots, Z_n$ conditionally independent.

Proof. Assume first that (23) holds. If $\mu = \sum_{r=1}^n \delta_{s_r}$ and Z_r are conditionally independent with laws $\mathcal{K}(s_r, \cdot)$, then

$$G_\mu[h] = \mathbb{E} \left[h^{\sum_{r=1}^n Z_r} \right] = \prod_{r=1}^n \mathbb{E}[h^{Z_r}].$$

The same product formula for $\mu + \eta$ immediately gives

$$G_{\mu+\eta}[h] = G_\mu[h]G_\eta[h].$$

Conversely, assume (22). Define

$$\mathcal{K}(s, \cdot) = \mathcal{Q}(\delta_s, \cdot).$$

The map $s \mapsto \delta_s$ is measurable for standard Borel S , so $\mathcal{K}$ is a Markov kernel. For $\mu = \sum_{r=1}^n \delta_{s_r}$, repeated use of (22) yields

$$G_\mu[h] = \prod_{r=1}^n G_{\delta_{s_r}}[h] = \prod_{r=1}^n \int h^\nu \mathcal{K}(s_r, d\nu).$$

The right-hand side is the generating functional of the convolution law $\mathcal{K}(s_1, \cdot) * \cdots * \mathcal{K}(s_n, \cdot)$. Probability laws on finite counting measures over a standard Borel space are determined by their generating, equivalently Laplace, functionals on bounded nonnegative measurable test functions; see, for example, standard random-measure uniqueness results [22]. Hence (23) follows. This proves the equivalence. $\square$

Remark 8.3 (Branching Markov kernels). *The property (22) is the Markov-kernel version of the branching property $\mathcal{Q}(\mu + \eta, \cdot) = \mathcal{Q}(\mu, \cdot) * \mathcal{Q}(\eta, \cdot)$. It is the continuous-state counterpart of Theorem 4.3. Spatial branching processes built from a Markov motion and a branching operator are treated in stochastic neutron-transport references such as Horton and Kyprianou [11].*

8.1 Mean and covariance measures

Assume the single-parent kernel $\mathcal{K}$ has finite second local moments, and assume either finite second total mass or enough local boundedness so that the covariance bimeasures below extend to finite signed measures on the sets considered. Define the mean offspring kernel

$$A(s, B) = \int_{\mathcal{N}_f(S)} \nu(B) \mathcal{K}(s, d\nu), \quad B \in \mathcal{S}, \quad (25)$$

and the single-parent covariance kernel

$$C_s(B, C) = \text{Cov}_{Z \sim \mathcal{K}(s, \cdot)}(Z(B), Z(C)). \quad (26)$$

For a random counting measure X_n , define the mean measure

$$\bar{\mu}_n(B) = \mathbb{E}[X_n(B)]$$

and covariance bimeasure

$$\Sigma_n(B, C) = \text{Cov}(X_n(B), X_n(C)).$$

When a genuine signed covariance measure is not needed, the formulas below may be read equivalently as identities for the covariance bilinear form evaluated on indicator test functions.

Proposition 8.4 (Continuous-state moment propagation). *For a continuous-state BSNN with finite second local moments,*

$$\bar{\mu}_{n+1}(B) = \int_S A(s, B) \bar{\mu}_n(ds), \quad (27)$$

and

$$\begin{aligned} \Sigma_{n+1}(B, C) &= \int_S C_s(B, C) \bar{\mu}_n(ds) \\ &\quad + \iint_{S \times S} A(s, B) A(t, C) \Sigma_n(ds, dt). \end{aligned} \quad (28)$$

Proof. Condition on $X_n = \sum_r \delta_{s_r}$. By independence of the offspring measures,

$$\mathbb{E}[X_{n+1}(B) \mid X_n] = \sum_r A(s_r, B) = \int_S A(s, B) X_n(ds).$$

Taking expectations gives (27). Similarly,

$$\text{Cov}(X_{n+1}(B), X_{n+1}(C) \mid X_n) = \sum_r C_{s_r}(B, C) = \int_S C_s(B, C) X_n(ds).$$

The law of total covariance then gives

$$\Sigma_{n+1}(B, C) = \int_S C_s(B, C) \bar{\mu}_n(ds) + \text{Cov} \left(\int A(s, B) X_n(ds), \int A(t, C) X_n(dt) \right),$$

which is (28). $\square$

For bounded measurable test functions φ , define the dual positive operator

$$(A\varphi)(s) = \int_S \varphi(r) A(s, dr). \quad (29)$$

Then

$$\mathbb{E}\langle \varphi, X_{n+1} \rangle = \mathbb{E}\langle A\varphi, X_n \rangle.$$

If $A(s, dr)$ has density $K(r, s) dr$, then the mean-density update has the familiar form

$$\phi_{n+1}(r) = \int_S K(r, s) \phi_n(s) ds. \quad (30)$$

This is the infinite-dimensional analogue of a positive linear recurrent network. In machine-learning language, it is closer to a positive linear neural operator than to a finite dense neural network. If the corresponding positive operator is compact and strongly positive on an appropriate Banach lattice, Krein–Rutman theory provides a positive principal eigenfunction and eigenvalue; neutron-transport settings often require more delicate positive-semigroup methods [10–12].

9 Finite-sample estimation guarantees

This section gives elementary finite-sample guarantees for estimating either the offspring law or the mean matrix, using Hoeffding-type concentration and standard sub-Gaussian net arguments [21, 23]. The strongest guarantees require parent-resolved observations: for each parent type i , one observes independent offspring vectors

$$Y_{i,1}, \dots, Y_{i,N_i} \sim Y_i.$$

This is the natural observation scheme for estimating the offspring PGFs themselves. Aggregate count trajectories are sufficient for estimating the conditional mean map under excitation/rank conditions, but not for nonparametrically identifying the full offspring PGFs without repeated transition-law information or additional assumptions.

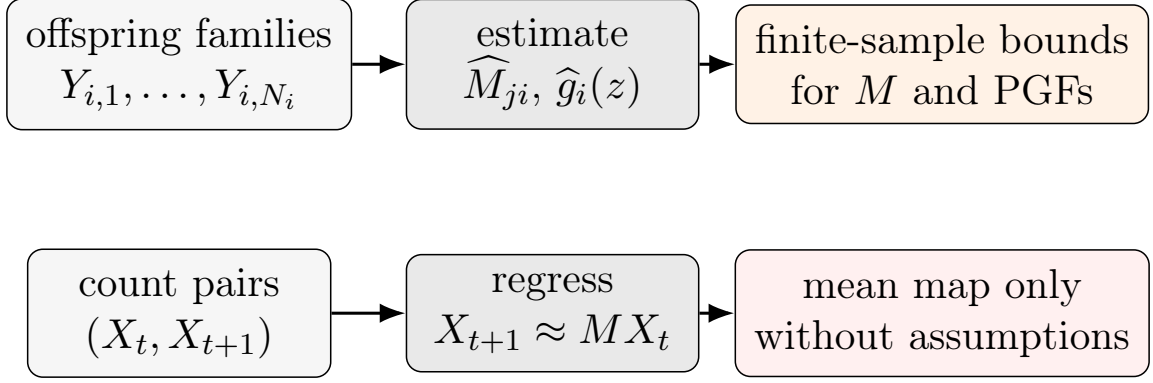


Figure 6: Observation regimes for statistical estimation. Parent-resolved family data can estimate one-parent PGFs, while aggregate trajectories generally estimate only the conditional mean matrix unless additional assumptions are imposed.

9.1 Parent-resolved estimation of M and PGFs

Define

$$\widehat{M}_{ji} = \frac{1}{N_i} \sum_{r=1}^{N_i} (Y_{i,r})_j, \quad \widehat{g}_i(z) = \frac{1}{N_i} \sum_{r=1}^{N_i} z^{Y_{i,r}}. \quad (31)$$

Let $N_{\min} = \min_i N_i$.

Theorem 9.1 (Finite-sample mean-matrix guarantee). *Assume that $(Y_i)_j \in [0, B]$ almost surely for every parent type i and child type j . Then, with probability at least $1 - \delta$,*

$$\max_{i,j} |\widehat{M}_{ji} - M_{ji}| \leq B \sqrt{\frac{\log(2m^2/\delta)}{2N_{\min}}}. \quad (32)$$

Consequently,

$$\|\widehat{M} - M\|_F \leq mB \sqrt{\frac{\log(2m^2/\delta)}{2N_{\min}}}. \quad (33)$$

Proof. For fixed (i, j) , Hoeffding's inequality gives

$$\mathbb{P} \left(|\widehat{M}_{ji} - M_{ji}| > \epsilon \right) \leq 2 \exp \left(-\frac{2N_i \epsilon^2}{B^2} \right) \leq 2 \exp \left(-\frac{2N_{\min} \epsilon^2}{B^2} \right).$$

Apply a union bound over the m^2 entries and solve for ϵ . The Frobenius bound follows from $\|A\|_F \leq m \max_{i,j} |A_{ji}|$. $\square$

Theorem 9.2 (Finite-sample PGF guarantee). *Let $\mathcal{Z} = \{z^{(1)}, \dots, z^{(L)}\} \subset [0, 1]^m$ be a finite grid. Then, with probability at least $1 - \delta$,*

$$\max_{i \leq m} \max_{\ell \leq L} |\widehat{g}_i(z^{(\ell)}) - g_i(z^{(\ell)})| \leq \sqrt{\frac{\log(2mL/\delta)}{2N_{\min}}}. \quad (34)$$

If, in addition, $\sum_j (Y_i)_j \leq B$ almost surely for all i , then for every $\alpha > 0$, with probability at least $1 - \delta$,

$$\max_i \sup_{z \in [0,1]^m} |\widehat{g}_i(z) - g_i(z)| \leq \alpha + \sqrt{\frac{\log(2m(1 + 4B/\alpha)^m/\delta)}{2N_{\min}}}. \quad (35)$$

Proof. For fixed i and z , the random variable z^{Y_i} lies in $[0, 1]$, so Hoeffding's inequality gives

$$\mathbb{P}(|\hat{g}_i(z) - g_i(z)| > \epsilon) \leq 2e^{-2N_i\epsilon^2}.$$

A union bound over mL pairs gives (34). For the uniform bound, note that if $\sum_j Y_j \leq B$ and $z, z' \in [0, 1]^m$, then the map $z \mapsto z^Y$ is B -Lipschitz in $\|\cdot\|_\infty$ because the sum of the absolute partial derivatives is bounded by $\sum_j Y_j \leq B$. Thus both g_i and $\hat{g}_i$ are B -Lipschitz. Apply the finite-grid bound to an $\alpha/(2B)$ -net of $[0, 1]^m$, whose cardinality is at most $(1 + 4B/\alpha)^m$ by the standard covering bound, and add at most α for the two Lipschitz approximation errors. $\square$

9.2 Aggregate count trajectories

The next guarantee is stated for independent one-step aggregate pairs, or for an exogenous design after conditioning on the design and retaining conditionally independent noise. A single long branching trajectory is adaptive because X_{t+1} also becomes part of the next design point; for that setting one should replace the conditional-independence assumption below by a martingale or mixing concentration argument. The sub-Gaussian assumption is also a modeling condition; unrestricted Poisson or supercritical count noise generally requires bounded/truncated observations or sub-exponential analogues. The point of the result is therefore identifiability and scaling of the aggregate mean estimator, not a sharp time-series theorem.

Suppose we observe one-step aggregate pairs (Φ_t, Y_t) , $t = 1, \dots, T$, satisfying

$$Y_t = M_\star \Phi_t + \varepsilon_t, \quad \mathbb{E}[\varepsilon_t | \Phi_t] = 0, \quad (36)$$

where $\Phi_t = X_t$ and $Y_t = X_{t+1}$. Let

$$G_T = \sum_{t=1}^T \Phi_t \Phi_t^\top, \quad \widetilde{M} = \left(\sum_{t=1}^T Y_t \Phi_t^\top \right) G_T^{-1}$$

when G_T is invertible.

Theorem 9.3 (Aggregate trajectory regression guarantee). *Condition on a realized design $\Phi_1, \dots, \Phi_T$ such that $\lambda_{\min}(G_T) \geq \gamma > 0$. Assume that, for each output coordinate j , the scalar noises $\varepsilon_{j,t}$ are conditionally independent and conditionally σ^2 -sub-Gaussian. Then, with conditional probability at least $1 - \delta$,*

$$\|\widetilde{M} - M_\star\|_F \leq 2\sigma \sqrt{\frac{2m}{\gamma} \log \left(\frac{2m5^m}{\delta} \right)}. \quad (37)$$

If $\widehat{M}_+$ is the projection of $\widetilde{M}$ onto the nonnegative cone in the prediction norm $\|A\|_{G_T}^2 = \text{tr}(AG_TA^\top)$, then the same right-hand side also bounds $\|\widehat{M}_+ - M_\star\|_F$ because $M_\star \geq 0$ is feasible.

Proof. For a fixed row j ,

$$\widetilde{M}_{j\cdot} - M_{\star,j\cdot} = \left(\sum_{t=1}^T \varepsilon_{j,t} \Phi_t^\top \right) G_T^{-1}.$$

Let $Z_j = G_T^{-1/2} \sum_t \varepsilon_{j,t} \Phi_t$. For any fixed unit vector u , conditional sub-Gaussianity gives

$$u^\top Z_j = \sum_t \varepsilon_{j,t} u^\top G_T^{-1/2} \Phi_t$$

as a σ^2 -sub-Gaussian variable, because

$$\sum_t (u^\top G_T^{-1/2} \Phi_t)^2 = u^\top G_T^{-1/2} G_T G_T^{-1/2} u = 1.$$

A standard 1/2-net argument on the unit sphere, using a net of size at most 5^m and the inequality $\|z\|_2 \leq 2 \sup_{u \in \mathcal{N}} |u^\top z|$ for a 1/2-net $\mathcal{N}$, yields

$$\|Z_j\|_2 \leq 2\sigma \sqrt{2 \log(2m5^m/\delta)}$$

simultaneously for all $j = 1, \dots, m$ with probability at least $1 - \delta$. Since

$$\|\widetilde{M}_j - M_{\star,j}\|_2 \leq \gamma^{-1/2} \|Z_j\|_2,$$

summing over j gives (37). For the projection claim, the same row-wise argument also gives $\|\widetilde{M} - M_\star\|_{G_T} \leq 2\sigma \sqrt{2m \log(2m5^m/\delta)}$. Projection onto a closed convex set cannot increase distance to a feasible point in the Hilbert norm $\|\cdot\|_{G_T}$, and $\|A\|_F \leq \gamma^{-1/2} \|A\|_{G_T}$, so the displayed Frobenius bound also holds for $\widehat{M}_+$. $\square$

Remark 9.4 (What can and cannot be estimated). *Parent-resolved observations estimate g_i and therefore the whole one-parent offspring law. Aggregate count pairs estimate the conditional mean map $x \mapsto Mx$ under rank conditions. Estimating the full offspring PGFs from aggregate trajectories alone requires richer repeated transition-law observations, parametric assumptions such as independent Poisson offspring, or direct family-level data. This distinction matters because many different offspring laws share the same mean matrix M but have different covariance and extinction behavior.*

10 Backpropagation, adjoints, and eigenvalue sensitivity

A neural-network contribution becomes more compelling if the network notation yields useful differentiation formulas. The mean BSNN does so directly.

Let M_θ be a differentiable family of nonnegative matrices parameterized by $\theta \in \mathbb{R}^p$, and consider the recurrence with fixed, θ -independent initial state x_0 :

$$x_{n+1} = M_\theta x_n, \quad n = 0, \dots, N-1. \quad (38)$$

Let the loss be

$$L(\theta) = \ell(x_N). \quad (39)$$

Theorem 10.1 (Backpropagation is adjoint sensitivity). *Define adjoint variables*

$$\lambda_N = \nabla_{x_N} \ell(x_N), \quad \lambda_n = M_\theta^\top \lambda_{n+1}, \quad n = N-1, \dots, 0. \quad (40)$$

Then, for each parameter coordinate θ_r ,

$$\frac{\partial L}{\partial \theta_r} = \sum_{n=0}^{N-1} \lambda_{n+1}^\top \left(\frac{\partial M_\theta}{\partial \theta_r} \right) x_n. \quad (41)$$

Proof. Differentiate the recurrence (38):

$$dx_{n+1} = M_\theta dx_n + (dM_\theta)x_n. \quad (42)$$

Since $dL = \lambda_N^\top dx_N$, recursively substitute (42) backward. The choice $\lambda_n = M_\theta^\top \lambda_{n+1}$ cancels the propagated state-variation terms and leaves

$$dL = \sum_{n=0}^{N-1} \lambda_{n+1}^\top (dM_\theta)x_n. \quad (43)$$

Setting $dM_\theta = (\partial M_\theta / \partial \theta_r) d\theta_r$ and collecting coefficients of $d\theta_r$ proves (41). $\square$

This theorem is exactly the algebra of backpropagation through a linear RNN. It is also the algebra of adjoint sensitivity analysis for a linear propagation operator. The BSNN formulation makes the equivalence explicit.

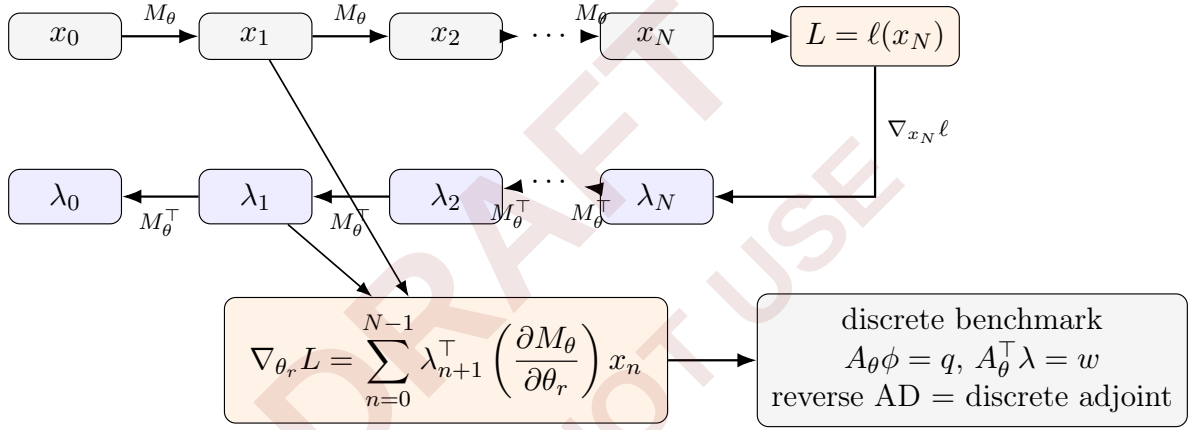


Figure 7: Backpropagation through the mean BSNN is the same vector-Jacobian algebra as a discrete adjoint. Forward states propagate by M_θ ; adjoints propagate backward by $M_\theta^\top$; gradients are accumulated from state-adjoint bilinear products.

A related formula applies to the dominant eigenvalue.

Theorem 10.2 (Dominant eigenvalue sensitivity). *Let M_θ be differentiable and non-negative. Suppose $k(\theta) = \rho(M_\theta)$ is a simple dominant eigenvalue with right and left eigenvectors v_θ and u_θ normalized by $u_\theta^\top v_\theta = 1$:*

$$M_\theta v_\theta = k(\theta) v_\theta, \quad u_\theta^\top M_\theta = k(\theta) u_\theta^\top. \quad (44)$$

Then

$$\frac{\partial k}{\partial \theta_r} = u_\theta^\top \left(\frac{\partial M_\theta}{\partial \theta_r} \right) v_\theta. \quad (45)$$

Proof. Differentiate $M_\theta v_\theta = k(\theta) v_\theta$ with respect to θ_r :

$$\left(\frac{\partial M_\theta}{\partial \theta_r} \right) v_\theta + M_\theta \frac{\partial v_\theta}{\partial \theta_r} = \frac{\partial k}{\partial \theta_r} v_\theta + k(\theta) \frac{\partial v_\theta}{\partial \theta_r}.$$

Left-multiply by $u_\theta^\top$. Since $u_\theta^\top M_\theta = k(\theta) u_\theta^\top$, the terms involving $\partial v_\theta / \partial \theta_r$ cancel. Using $u_\theta^\top v_\theta = 1$ gives (45). $\square$

Remark 10.3 (Differentiable positive parameterizations). *For learning or inverse problems, one can enforce nonnegativity by parameterizing*

$$M_\theta = \text{softplus}(A_\theta)$$

entrywise, or by using another positive parameterization. Stability or criticality-aware regularizers may penalize $\rho(M_\theta)$, for example

$$R_0(\theta) = (\log \rho(M_\theta))^2, \quad R_1(\theta) = \max\{0, \rho(M_\theta) - 1\}^2.$$

These are mathematical suggestions for synthetic or abstract data experiments, not operational prescriptions for physical systems.

10.1 Fully worked discrete-adjoint benchmark

To make the equivalence between automatic differentiation and a classical discrete adjoint concrete, consider the following synthetic positive operator benchmark. It is a dimensionless linear diffusion/transport-like solve, not a reactor model. Let $D \in \mathbb{R}^{(N-1) \times N}$ be the first-difference matrix, let

$$A_\theta = \text{diag}(a_\theta) + D^\top \text{diag}(d_\theta) D, \quad a_\theta = a_0 + \text{softplus}(\alpha), \quad d_\theta = d_0 + \text{softplus}(\beta), \quad (46)$$

where $a_0, d_0 > 0$. Given a source vector q and response vector w , solve

$$A_\theta \phi = q, \quad J(\theta) = w^\top \phi. \quad (47)$$

Proposition 10.4 (Reverse AD equals the discrete adjoint). *Let λ solve the adjoint system*

$$A_\theta^\top \lambda = w. \quad (48)$$

For the benchmark (46)–(47), the gradient is

$$\frac{\partial J}{\partial \alpha_i} = -\lambda_i \phi_i \sigma(\alpha_i), \quad (49)$$

where $\sigma(t) = 1/(1 + e^{-t})$ is the derivative of softplus, and

$$\frac{\partial J}{\partial \beta_e} = -(D\lambda)_e (D\phi)_e \sigma(\beta_e). \quad (50)$$

These are exactly the vector-Jacobian products returned by reverse-mode automatic differentiation through the linear solve.

Proof. Differentiate $A_\theta \phi = q$ with fixed q :

$$A_\theta d\phi + (dA_\theta)\phi = 0.$$

Since $A_\theta^\top \lambda = w$,

$$dJ = w^\top d\phi = \lambda^\top A_\theta d\phi = -\lambda^\top (dA_\theta)\phi.$$

For an absorption parameter α_i , $\partial A_\theta / \partial \alpha_i = \sigma(\alpha_i) e_i e_i^\top$, which gives (49). For an edge coefficient β_e , $\partial A_\theta / \partial \beta_e = \sigma(\beta_e) D^\top e_e e_e^\top D$, which gives (50). Reverse-mode automatic differentiation through the computational graph for $\phi = A_\theta^{-1} q$ applies the same transpose linear solve and therefore yields the same formulas. This is the finite-dimensional algebra underlying discrete adjoint transport sensitivity calculations; continuous/discrete adjoint equivalence for multigroup neutron diffusion has been demonstrated in reactor-physics settings by Wu et al. [24]. $\square$

A reproducible run with $N = 10$ synthetic cells, fixed seed 11, and double precision gave the verification numbers in Table 2. The maximum absolute discrepancy between the closed-form adjoint and automatic differentiation was at roundoff level.

Table 2: Synthetic discrete-adjoint benchmark. AD denotes reverse-mode automatic differentiation through the discretized solve. FD denotes central finite difference checks at selected coordinates.

Quantity	Value	Interpretation
$J = w^\top \phi$	6.5122621426	response functional
$\max_i g_{\alpha_i}^{\text{adj}} - g_{\alpha_i}^{\text{AD}} $	5.6×10^{-17}	absorption-gradient agreement
$\max_e g_{\beta_e}^{\text{adj}} - g_{\beta_e}^{\text{AD}} $	7.9×10^{-17}	diffusion-gradient agreement
Largest listed FD-adjoint error	6.6×10^{-10}	finite-difference verification

11 Nonnegative recurrent architectures and regularization

The BSNN viewpoint suggests recurrent architectures that are intentionally less expressive than generic RNNs but better aligned with branching data. The purpose is not universal approximation. It is to preserve nonnegativity, a valid branching transition law, interpretable extinction probabilities, covariance recursions, and spectral-radius control.

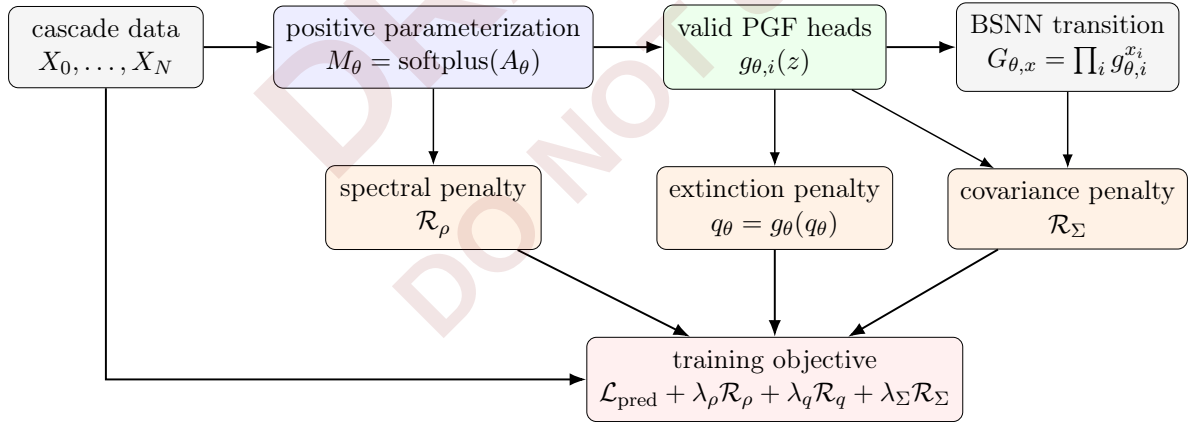


Figure 8: Structure-aware BSNN training. Nonnegative parameterizations preserve positive mean dynamics, PGF heads preserve valid branching transition laws, and regularizers can target spectral radius, extinction behavior, and covariance propagation.

Positive mean architecture

The minimal mean architecture is

$$x_{n+1} = M_\theta x_n, \quad M_\theta = \text{softplus}(A_\theta) + \epsilon \mathbf{1}\mathbf{1}^\top, \quad (51)$$

with $\epsilon \geq 0$. This enforces $M_\theta \geq 0$. A normalized version can directly control the Perron–Frobenius gain:

$$M_\theta = \alpha \frac{\widetilde{M}_\theta}{\rho(\widetilde{M}_\theta) + \varepsilon}, \quad \widetilde{M}_\theta = \text{softplus}(A_\theta), \quad (52)$$

so that $\rho(M_\theta) \approx \alpha$ when ε is small.

Valid offspring-law heads

A mean matrix does not determine a BSNN law. One can instead parameterize the one-parent PGFs. Three useful heads are:

1. Independent Poisson head:

$$g_{\theta,i}(z) = \exp\left(\sum_{j=1}^m M_{\theta,ji}(z_j - 1)\right), \quad V_{\theta,i} = \text{diag}(M_{\theta,1i}, \dots, M_{\theta,mi}). \quad (53)$$

2. Overdispersed negative-binomial head:

$$g_{\theta,i}(z) = \prod_{j=1}^m \left(\frac{p_{\theta,ji}}{1 - (1 - p_{\theta,ji})z_j} \right)^{r_{\theta,ji}}, \quad (54)$$

where $r_{\theta,ji} > 0$ and $p_{\theta,ji} \in (0, 1)$ are neural outputs. This separates mean and variance.

3. Finite-support empirical PGF head:

$$g_{\theta,i}(z) = \sum_{y \in \mathcal{Y}_i} \pi_{\theta,i}(y) z^y, \quad \pi_{\theta,i}(y) \geq 0, \quad \sum_{y \in \mathcal{Y}_i} \pi_{\theta,i}(y) = 1. \quad (55)$$

The full transition PGF is

$$G_{\theta,x}(z) = \prod_{i=1}^m g_{\theta,i}(z)^{x_i}. \quad (56)$$

Proposition 11.1 (Validity of PGF architectures). *If each $g_{\theta,i}$ is a probability-generating function on $\mathbb{N}_0^m$, then (56) defines a BSNN transition law. Its mean matrix is*

$$M_{\theta,ji} = \left. \frac{\partial g_{\theta,i}(z)}{\partial z_j} \right|_{z=\mathbf{1}}. \quad (57)$$

Proof. For each i , let Y_i have PGF $g_{\theta,i}$. The product (56) is the PGF of the sum of x_i independent copies of Y_i for each type i . Theorem 4.3 implies that the transition law is a BSNN. Differentiating the one-parent PGF at $z = \mathbf{1}$ gives the first moment. $\square$

Extinction, covariance, and spectral-radius penalties

A structure-aware training objective can be written as

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{pred}}(\theta) + \lambda_\rho \mathcal{R}_\rho(\theta) + \lambda_q \mathcal{R}_q(\theta) + \lambda_\Sigma \mathcal{R}_\Sigma(\theta). \quad (58)$$

For a target gain k_0 or a maximum permitted gain $k_{\max}$,

$$\mathcal{R}_\rho(\theta) = (\log \rho(M_\theta) - \log k_0)^2, \quad \mathcal{R}_\rho^+(\theta) = \max\{0, \rho(M_\theta) - k_{\max}\}^2. \quad (59)$$

The extinction vector q_θ is the minimal fixed point

$$q_\theta = g_\theta(q_\theta), \quad g_\theta = (g_{\theta,1}, \dots, g_{\theta,m}). \quad (60)$$

If synthetic survival/extinction labels or empirical extinction frequencies $\hat{q}$ are available, one may use

$$\mathcal{R}_q(\theta) = \|q_\theta - \hat{q}\|_2^2 \quad (61)$$

or the corresponding Bernoulli survival likelihood.

For covariance matching, let $\hat{x}_n$ and $\hat{\Sigma}_n$ be empirical means and covariances across trajectories. Let $V_{\theta,i}$ be the one-parent offspring covariance from the PGF head. Define

$$\Delta_{\theta,n} = \hat{\Sigma}_{n+1} - M_\theta \hat{\Sigma}_n M_\theta^\top - \sum_{i=1}^m \hat{x}_{n,i} V_{\theta,i}, \quad (62)$$

and set

$$\mathcal{R}_\Sigma(\theta) = \sum_n \|\Delta_{\theta,n}\|_F^2. \quad (63)$$

This penalty forces the learned stochastic recurrent network to match not only one-step means but also the branching-noise recursion in Theorem 6.1.

Proposition 11.2 (Differentiating extinction penalties). *Suppose g_θ is continuously differentiable and $q_\theta \in (0, 1)^m$ solves (60). If $I - D_q g_\theta(q_\theta)$ is invertible, then q_θ is locally differentiable and*

$$\frac{\partial q_\theta}{\partial \theta_r} = (I - D_q g_\theta(q_\theta))^{-1} \frac{\partial g_\theta(q_\theta)}{\partial \theta_r}. \quad (64)$$

Proof. Let $F(q, \theta) = q - g_\theta(q)$. Then $F(q_\theta, \theta) = 0$ and

$$D_q F(q_\theta, \theta) = I - D_q g_\theta(q_\theta).$$

The implicit function theorem gives local differentiability. Differentiating $q_\theta - g_\theta(q_\theta) = 0$ with respect to θ_r and rearranging gives (64). $\square$

Remark 11.3 (Why these regularizers are not generic RNN regularizers). *The spectral term controls mean growth, the extinction term controls sample-path survival behavior, and the covariance term controls branching fluctuations. A generic RNN trained only for one-step prediction need not satisfy any of these three constraints.*

11.1 Explicit three-cell adjoint calculation

For a fully explicit hand-checkable benchmark, take

$$A_\theta = \begin{pmatrix} 2+\theta & -1 & 0 \\ -1 & 2+\theta & -1 \\ 0 & -1 & 2+\theta \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad c = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad J(\theta) = c^\top A_\theta^{-1} b. \quad (65)$$

This is a dimensionless positive-response linear solve, not a physical reactor calculation. At $\theta = 1/2$,

$$\phi = A_\theta^{-1} b = \frac{1}{85} \begin{pmatrix} 42 \\ 20 \\ 8 \end{pmatrix}, \quad J = \frac{8}{85}. \quad (66)$$

The adjoint equation $A_\theta^\top \psi = c$ gives

$$\psi = \frac{1}{85} \begin{pmatrix} 8 \\ 20 \\ 42 \end{pmatrix}. \quad (67)$$

Since $\partial A_\theta / \partial \theta = I$, the adjoint formula gives

$$\frac{dJ}{d\theta} = -\psi^\top \phi = -\frac{1072}{7225} \approx -0.1483737024. \quad (68)$$

Centered finite differences give -0.1483738706 at $h = 10^{-3}$, -0.1483737024 at $h = 10^{-5}$, and -0.1483737021 at $h = 10^{-7}$. Thus a reverse-mode automatic-differentiation calculation through the solve, the classical discrete adjoint formula, and finite differences agree on this benchmark.

12 Expressivity limitations

The BSNN formulation is useful only if its limitations are explicit.

Theorem 12.1 (Mean expressivity limitation). *Every mean BSNN map*

$$F(x) = Mx, \quad M \geq 0, \quad (69)$$

is positive homogeneous, additive, and monotone on $\mathbb{R}_+^m$. That is, for $x, y \in \mathbb{R}_+^m$ and $\alpha, \beta \geq 0$,

$$F(\alpha x + \beta y) = \alpha F(x) + \beta F(y), \quad (70)$$

while

$$x \leq y \implies F(x) \leq F(y) \quad (71)$$

componentwise. Therefore mean BSNN maps cannot represent arbitrary neural-network layers or arbitrary nonlinear functions.

Proof. Equation (70) follows from matrix linearity:

$$M(\alpha x + \beta y) = \alpha Mx + \beta My.$$

If $x \leq y$, then $y - x \geq 0$. Since $M \geq 0$, $M(y - x) \geq 0$, so $My - Mx \geq 0$, proving (71). A nonlinear function such as $g(t) = t^2$ on $[0, 1]$ cannot be represented by such a linear map, since

$$g(1/2) = 1/4 \neq 1/2 = \frac{1}{2}g(1) + \frac{1}{2}g(0).$$

Thus the class is not universal for nonlinear function approximation. $\square$

Proposition 12.2 (Law-level limitation). *A count-valued stochastic recurrent network is a BSNN only if its transition PGF factorizes as*

$$G_x(z) = \prod_{i=1}^m g_i(z)^{x_i}. \quad (72)$$

Consequently, BSNNs exclude stochastic networks with same-layer interaction, competition, saturation, attention, normalization, or gating among existing units unless those mechanisms are included by augmenting the state and preserving conditional branching.

Proof. The factorization follows from Theorem 4.3. Mechanisms such as saturation or competition generally make the offspring law of one unit depend on the number or types of other units in the same generation. Then the transition law from $x + y$ is not the convolution of the laws from x and y , so $G_{x+y} \neq G_x G_y$ in general. Hence such systems lie outside the BSNN class unless the state representation is enlarged so that conditional independence is restored. $\square$

Remark 12.3 (What the analogy gets wrong). *A physical fission chain has no training loss, no learned semantic objective, and no arbitrary negative weights or learned nonlinear activations. Feedbacks such as material changes, control actions, and delayed effects can make an effective operator state-dependent, e.g.*

$$x_{n+1} = M(\eta_n)x_n,$$

where η_n represents environmental or material state. Such models are controlled or state-dependent positive stochastic systems, not fixed-weight feed-forward neural networks.

13 Random computation graph structure

Theorem 13.1 (A realized BSNN history is a random layered DAG). *For any finite horizon N , a realized BSNN history defines a directed acyclic graph $G_N = (V, E)$ layered by generation. Thus it is a stochastic feed-forward computation graph of random width.*

Proof. Let V_n be the set of realized units in generation n , and set $V = \bigcup_{n=0}^N V_n$. Draw an edge from a unit $a \in V_n$ to a unit $b \in V_{n+1}$ when b is an offspring of a . Every directed edge increases the generation index by one. A directed cycle would require the generation index to strictly increase along the cycle and then return to its starting value, which is impossible. Hence G_N is a directed acyclic graph. The cardinalities $|V_n|$ are random, so the graph has random width. $\square$

14 Synthetic comparison with unconstrained recurrent training

This section reports a small synthetic benchmark. It is included to demonstrate how the BSNN theory can be tested, not to claim broad empirical superiority. Figure 9 summarizes the comparison workflow. The data are dimensionless cascade counts generated from a three-type Poisson BSNN:

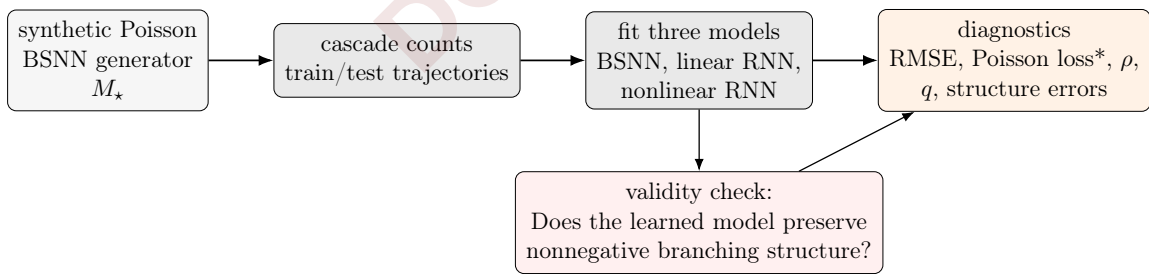


Figure 9: Synthetic comparison workflow. The benchmark tests whether constrained BSNN training preserves branching structure and spectral/extinction diagnostics compared with unconstrained recurrent predictors.

$$X_{n+1,j} \mid X_n \sim \text{Poisson}((M_\star X_n)_j), \quad j = 1, 2, 3, \quad (73)$$

independently across j , with

$$M_\star = 1.08 \frac{B}{\rho(B)}, \quad B = \begin{pmatrix} 0.55 & 0.10 & 0.05 \\ 0.25 & 0.60 & 0.10 \\ 0.05 & 0.20 & 0.50 \end{pmatrix}. \quad (74)$$

Thus $\rho(M_\star) = 1.08$. The training set contains 30 trajectories and the test set contains 300 trajectories. Three models are fit: a BSNN mean model trained by row-wise nonnegative least squares; an unconstrained linear RNN with bias trained by least squares; and an unconstrained nonlinear recurrent cell using a one-hidden-layer random-feature map with least-squares output training. The first model is branching-valid at the mean level; the latter two are prediction models that do not define BSNN transition laws without extra constraints.

Table 3: Synthetic cascade benchmark. The data are generated from the dimensionless three-type Poisson BSNN in (74), using 30 training trajectories and 300 test trajectories. Lower is better except for the validity column. The Poisson loss* omits the log-factorial constants, so it can be negative.

Model	RMSE	Pois. loss*	$ \Delta\rho $	Rel. M	Ext. err.	Valid?
BSNN, nonnegative LS	6.457	-117.284	0.014	0.071	0.049	yes
Unconstrained linear RNN	6.479	-117.281	0.042	0.085	–	no
Unconstrained nonlinear RNN	102.216	-103.099	–	–	–	no

Table 4: Structural diagnostics for the same synthetic benchmark. Homogeneity and additivity errors measure violations of the branching mean law. Negative mass is $\sum_{ij} \max\{-\hat{M}_{ij}, 0\}$.

Model	Homog. err.	Add. err.	Neg. mass	$\hat{\rho}$
BSNN, nonnegative LS	0.000	0.000	0.000	1.066
Unconstrained linear RNN	0.037	0.037	0.000	1.038
Unconstrained nonlinear RNN	1.416	0.775	–	–

The constrained BSNN and the unconstrained linear RNN have similar one-step predictive error, as expected because the data-generating conditional mean is linear. The BSNN estimate has smaller spectral-radius error and smaller relative mean-matrix error in this run. It also has an interpretable extinction-vector error because its fitted transition law is branching-valid; the unconstrained recurrent predictors do not define extinction probabilities without adding a post hoc PGF model. More importantly, the BSNN exactly preserves the homogeneity and additivity identities required by the branching mean law. The nonlinear unconstrained cell has higher prediction error in this low-data setting and strongly violates branching structure. These results should be read as a reproducibility smoke test, not as a comprehensive empirical study.

15 Discussion

The BSNN formulation is mainly a bridge. For neutron-transport specialists, it packages a finite-type branching process as a stochastic recurrent computation graph. For machine-learning researchers, it identifies a physically motivated class of positive, count-valued stochastic networks with exact moment, extinction, and spectral theory.

A paper submitted to a neural-network venue should probably emphasize the general BSNN class rather than fission itself. The fission-chain model can then serve as a

motivating instance of a broader family of branching computation graphs. The publishable ML contribution would be strongest if accompanied by training algorithms, stability regularization, expressivity theorems, and reproducible synthetic benchmarks.

A paper submitted to a nuclear-engineering or applied-probability venue should emphasize the opposite direction: the neural-network notation is useful only insofar as it clarifies moment propagation, adjoint sensitivities, and the relationship between branching fluctuations and mean-field eigenvalue calculations. In that setting, the contribution must be positioned carefully against the existing stochastic neutron transport literature.

16 Conclusion

Discretized fission multiplication can be represented rigorously as a branching stochastic neural network. The exact equivalence is not that fission is a general artificial neural network, but that its generation-to-generation transition law is a count-valued stochastic recurrent network satisfying a branching PGF factorization. The mean dynamics are the positive linear RNN

$$x_{n+1} = Mx_n,$$

while the full stochastic dynamics retain covariance growth, extinction probabilities, and random genealogy. Criticality corresponds to the Perron–Frobenius spectral radius of M , and backpropagation through the mean model is equivalent to an adjoint sensitivity calculation.

The analogy is therefore scientifically defensible only in a limited form:

fission multiplication is an instance of positive branching stochastic recurrent computation.

It is not a trained, universal, semantic neural network. That limitation is not a weakness of the result; it is the condition that makes the result precise.

A One-type extinction proof

Consider the one-type process

$$X_{n+1} = \sum_{r=1}^{X_n} \xi_r^{(n)}, \quad (75)$$

where ξ is an $\mathbb{N}_0$ -valued offspring random variable with PGF

$$f(s) = \mathbb{E}[s^\xi] \quad (76)$$

and mean $\mu = f'(1) = \mathbb{E}[\xi]$. Let q be the extinction probability starting from $X_0 = 1$.

Proposition A.1 (One-type extinction criterion). *The extinction probability q is the smallest solution of*

$$f(q) = q \quad (77)$$

in $[0, 1]$. If the process is nondegenerate, then

$$\mu \leq 1 \implies q = 1, \quad \mu > 1 \implies q < 1. \quad (78)$$

Proof. Let $q_n = \mathbb{P}(X_n = 0 \mid X_0 = 1)$. Extinction by generation $n + 1$ occurs precisely when every first-generation child has a lineage extinct by generation n . Hence

$$q_{n+1} = \sum_{r=0}^{\infty} \mathbb{P}(\xi = r) q_n^r = f(q_n), \quad q_0 = 0.$$

The sequence (q_n) is nondecreasing and bounded above by 1, so $q_n \rightarrow q \in [0, 1]$. By continuity of f , $q = f(q)$. If $r \in [0, 1]$ is any fixed point, induction gives $q_n \leq r$ for all n , and therefore $q \leq r$. Thus q is the smallest fixed point.

The function f is convex on $[0, 1]$, satisfies $f(1) = 1$, and has $f'(1) = \mu$. If $\mu \leq 1$, convexity implies that no fixed point strictly below 1 exists, except in degenerate cases that are excluded by the nondegeneracy assumption; hence $q = 1$. If $\mu > 1$, then for $s < 1$ sufficiently close to 1, convexity and $f'(1) > 1$ imply $f(s) < s$. Since $f(0) \geq 0$, continuity implies the existence of a fixed point in $(0, 1)$. The minimal fixed point is therefore less than 1. $\square$

B Additional proof of the product form

For completeness, we restate the central product identity in coordinates. Let G_x be the transition PGF of a BSNN. Since

$$x = (x_1, \dots, x_m) = x_1 e_1 + \dots + x_m e_m,$$

branching factorization gives

$$G_x(z) = G_{e_1}(z)^{x_1} \dots G_{e_m}(z)^{x_m}.$$

Taking first derivatives at $z = \mathbf{1}$ gives the mean matrix:

$$M_{ji} = \left. \frac{\partial G_{e_i}(z)}{\partial z_j} \right|_{z=\mathbf{1}}.$$

Taking second derivatives gives second factorial moments, from which covariance matrices V_i can be reconstructed. Thus the neural-network weights and branching-noise terms are encoded directly in the one-parent PGFs.

C Reproducibility notes and remaining limitations

The five reviewer-facing additions listed in the previous draft have been incorporated into the manuscript: the continuous-state Markov-kernel theorem appears in Section 8; finite-sample estimation bounds appear in Section 9; the synthetic BSNN-versus-RNN comparison appears in Section 14; nonnegative recurrent architectures and regularizers appear in Section 11; and the worked automatic-differentiation/discrete-adjoint benchmark appears in Section 10.1.

Important limitations remain. First, the continuous-state theorem is a characterization of discrete-generation branching kernels on finite counting measures; it is not a full continuous-time transport semigroup theory. Second, the finite-sample guarantees are intentionally elementary and rely on either parent-resolved offspring observations or aggregate trajectories satisfying excitation and sub-Gaussian noise assumptions. Third,

the experiments are synthetic and demonstrate structural recovery, not superiority on operational reactor-physics data. Fourth, the architecture section gives safe, abstract, nonnegative recurrent models; it does not prescribe physical design choices. These limitations should be kept explicit in any submitted version.

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